

Motivation & Problem

CHALLENGE

Lunar rovers need absolute attitude for navigation, but **cannot rely on GNSS or a magnetic reference**.

LIMITATION

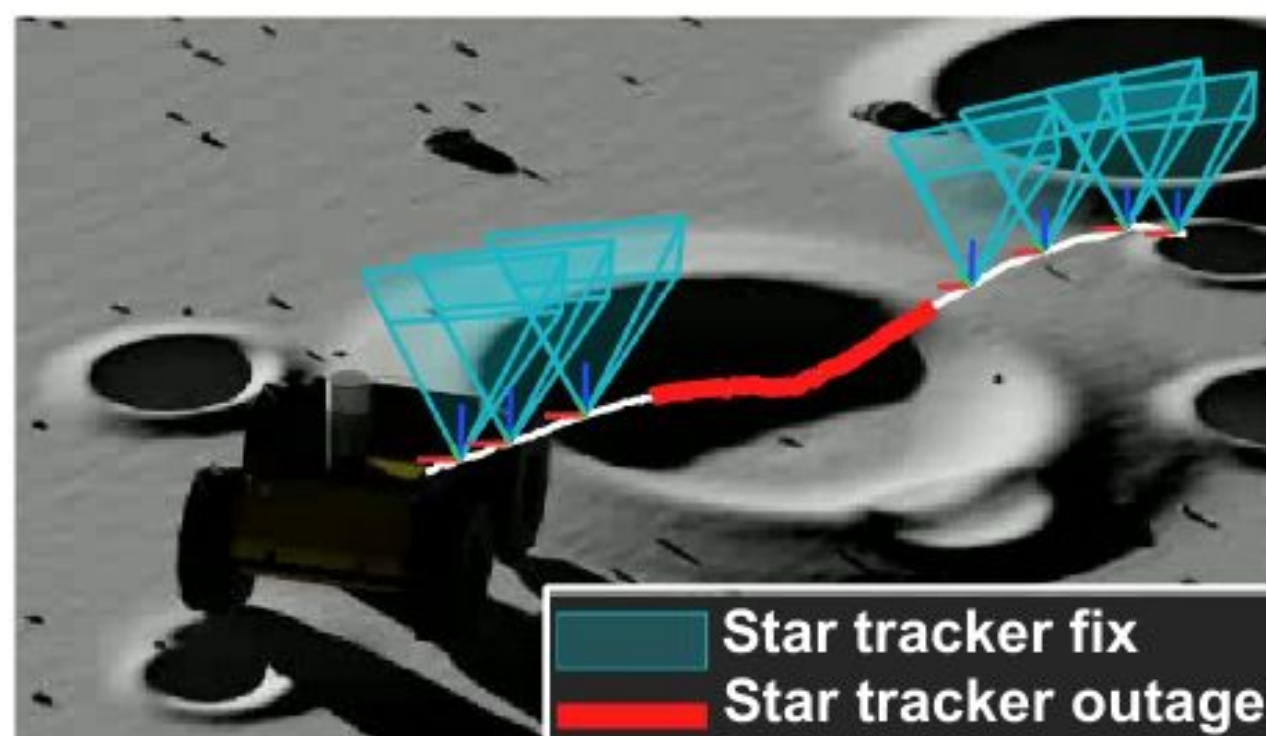
- Star trackers and Star-ID: easily fail during fast motion.
- Relative tracking: Accumulates drift over time.

OUR GOAL

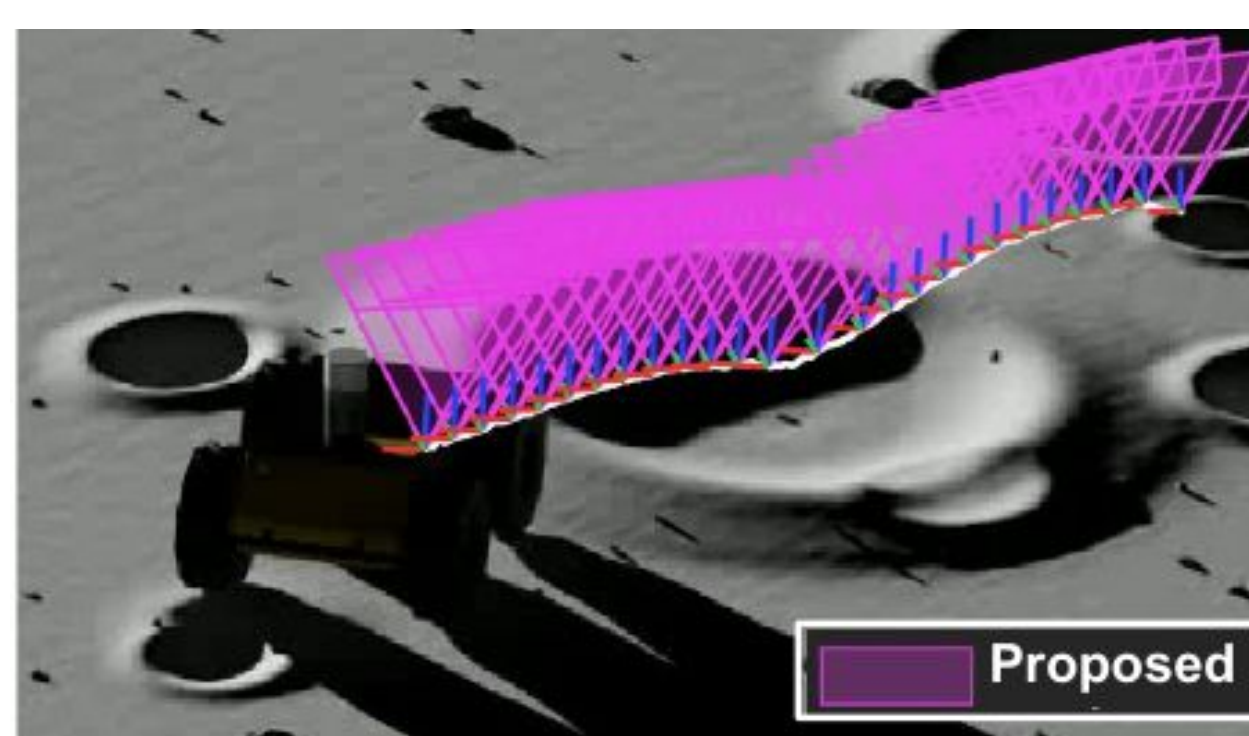
Use **relative tracking** to bridge output gaps and **catalog re-anchoring** to limit drift → **Maintain continuous, accurate absolute attitude estimates during fast rover motion**.

Continuous tracking. Drift correction. One camera.

STAR TRACKER

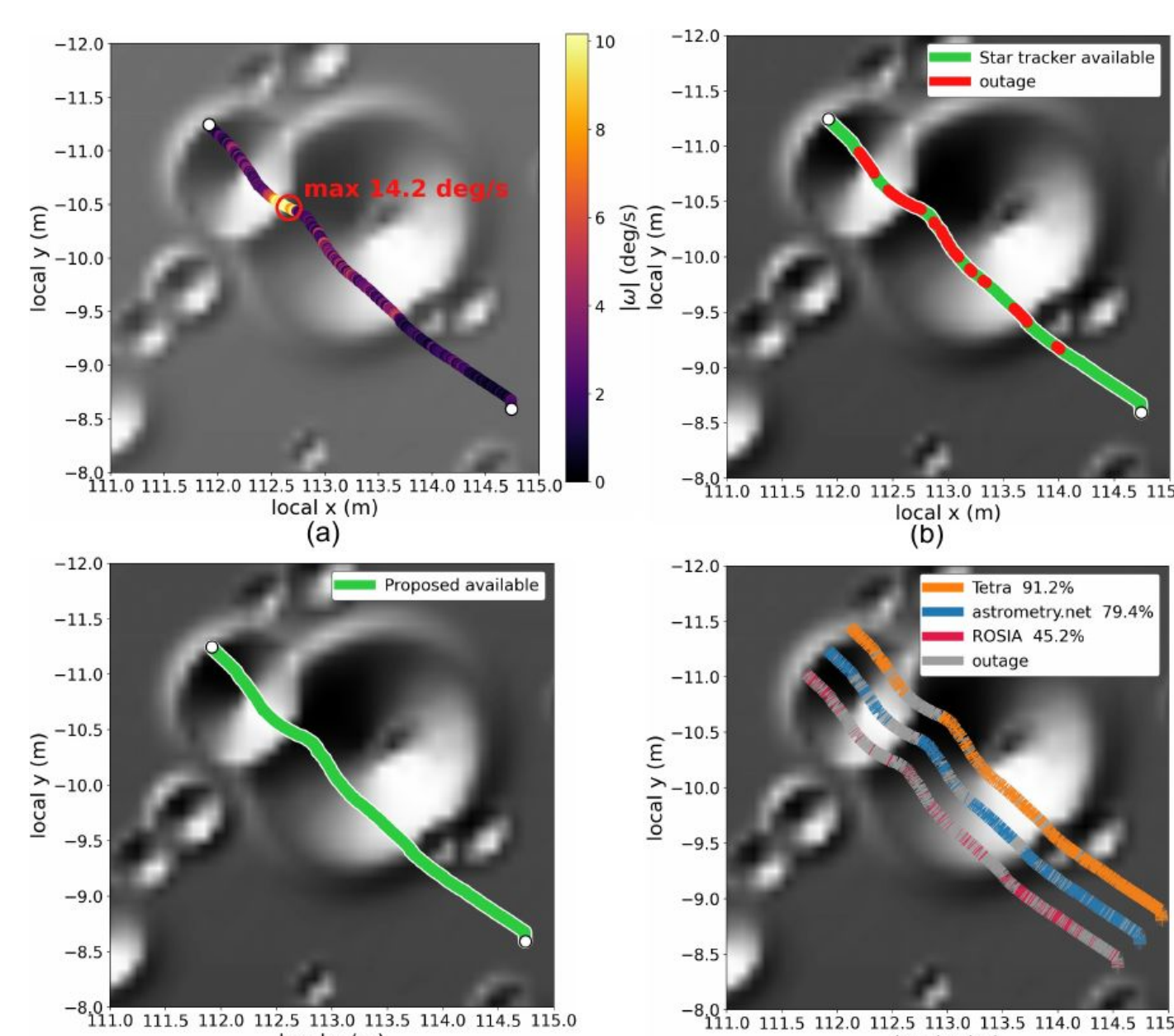
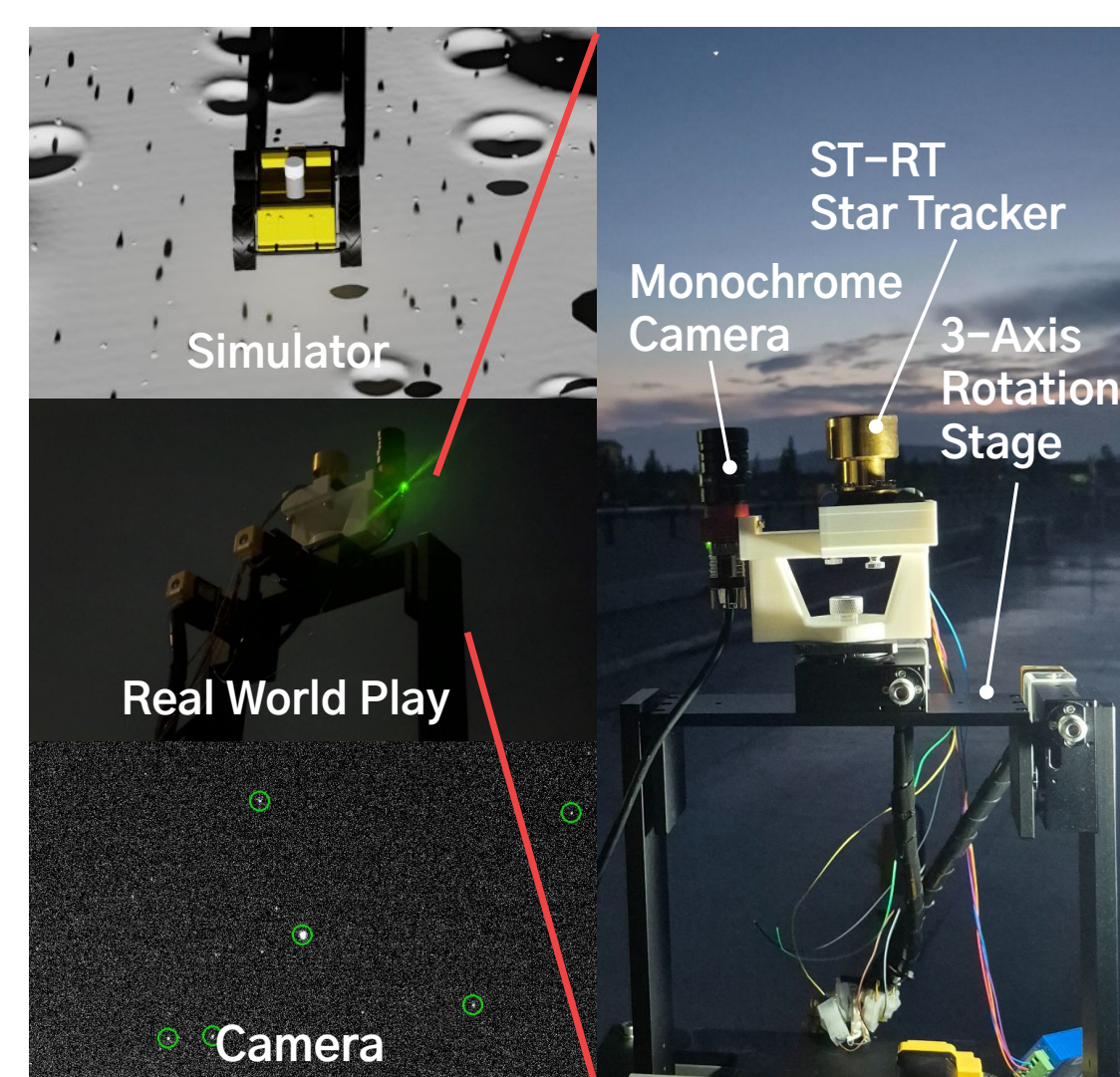


PROPOSED



Evaluation & Conclusion

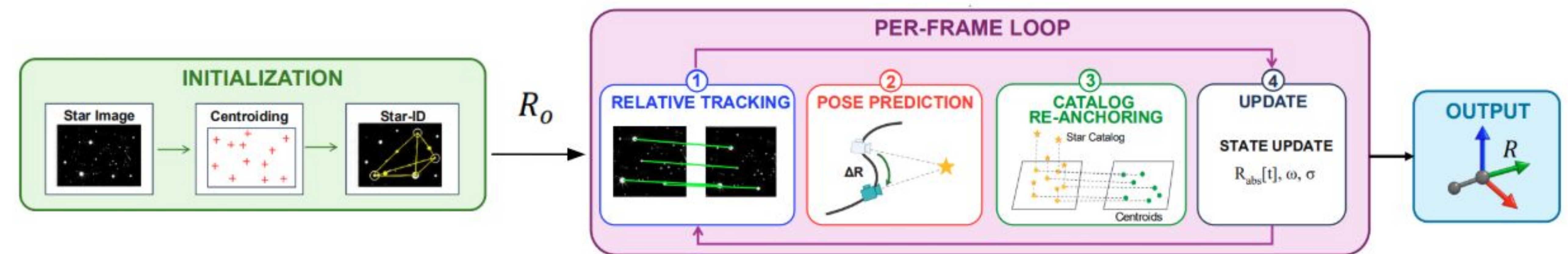
Spatial output availability



- Rapid motion:** Sharp attitude changes near crater boundaries
- Star tracker / Star-ID: Output gaps during fast rover motion
- Proposed: **Continuous attitude estimates along the full rover trajectory**

Proposed Method

- The pipeline takes star images, camera calibration, and a star catalog as input.
- Each frame produces an attitude estimate, marked as catalog-confirmed or propagated.



① RELATIVE TRACKING

Match stars with **LightGlue** and select RANSAC inliers $\mathcal{F}$. Solve **Wahba's problem** to estimate the relative rotation ΔR_t .

$$\Delta R_t = \arg \min_{R \in \text{SO}(3)} \sum_{i \in \mathcal{I}} \|\mathbf{b}_i^t - R \mathbf{b}_i^{t-1}\|^2$$

② POSE PREDICTION

Apply the relative rotation ΔR_t to the previous attitude R_{t-1} . The predicted attitude R_t^- gives the current camera orientation.

$$R_t^- = \Delta R_t R_{t-1}$$

③ CATALOG RE-ANCHORING

Project catalog stars $\mathbf{s}_i$ using R_t^- to obtain $\hat{\mathbf{u}}_{i,t}$. Match these projections to detected centroids and refine a candidate absolute attitude R_t^{ref} . $\hat{\mathbf{u}}_{i,t} = \pi(R_t^- \mathbf{s}_i)$

④ QUALITY-GATED UPDATE

Check reprojection and rotation consistency of R_t^{ref} . Set R_t to R_t^{ref} if valid, or retain R_t^- otherwise.

RESULTS & TAKEAWAY

Relative tracking + re-anchoring

Track rotation between frames and correct drift using the star catalog.

Continuous tracking

Robust during fast motion

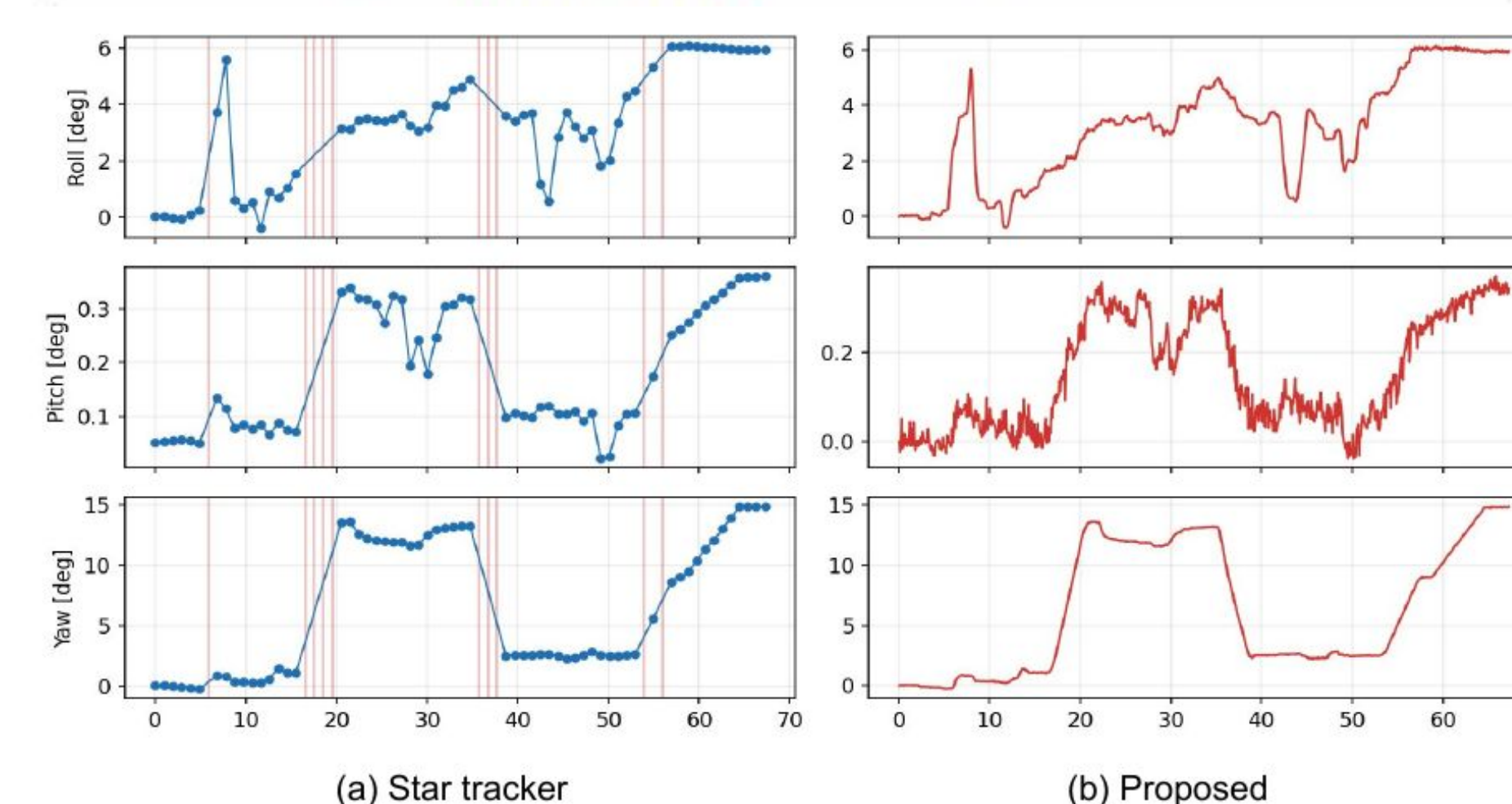
Performance across rover motions

0.147°–0.439° RMSE across six sequences

Continuous, robust attitude estimation for rover localization from a single camera.

Attitude RMSE & Availability

Rover	Sequence	RMSE [°]	ST avail. [%]
Husky	A (waypoint)	0.269	81.1
	B (straight)	0.159	83.3
	C (long)	0.439	87.2
	D (zigzag)	0.313	87.7
Pragyaan	E (zigzag)	0.257	77.6
	F (straight, 0.2 m/s)	0.147	91.6



- Attitude RMSE below 0.44°** across all six sequences.

- Our method achieves **100% output availability** across six sequences, compared with 84.8% on average for the star tracker.